

Evidence from the European Football Market

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Roadmap

- 1 Motivation & Background
- 2 Mirrleesian Framework
- 3 Institutional Framework
- 4 Kleven, Landais, Saez - Introduction
- 5 Theoretical Model
- 6 Reduced Form Results
- 7 Empirical Estimation

Question & Motivation

- **Question:** How do favorable tax policies induce migration for high-skilled individuals?
- Using data which tracks footballers' movement across leagues and countries, how did tax policies in Spain and Denmark affect migration decisions?
- Find notable shifts in demographic makeups of national leagues following policy changes, indicating that favorable policy leads to an influx of high skilled labor.

Why Football?

- Football contracts and by extension, movements, are easily tracable given it is a profession which is for entertainment
- Footballers at an elite level are certainly in the top end of the income distribution, thus leading them to be directly impacted by a tax cut for high skilled labor
- Football specific policy (Bosman Rule) provides another source of mobility variation, allowing for analysis of tax policy at multiple phases in the "football life cycle"

Contribution

- **What's new:** Implements a novel dataset of high-earning, high-skilled laborers' mobility patterns
- **Why it matters:** Allows for analysis of cross-country migration in the face of tax policy changes

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Overview

- 1981 Paper: "Migration and Optimal Income Taxes"
- How to optimally tax citizens earning income abroad as to reduce waste
- How do taxes impact the distribution of agents?
- How do taxes play as incentives for agents?

Model Considerations

The three criteria concerning migration research, according to Mirrlees

- ① Restriction to the group who does not migrate
 - Suboptimal as this does not account for those who wished to move but were constrained
 - Misses those who leave and still contribute to their country, a group which countries do not appear to be too concerned with
 - Restriction of taxes and corresponding benefits to voters, a confusing group as there are various definitions of voters
- ② Definition of "Nationals"
 - Overly restrictive and presumptuous (i.e. exclusion of immigrants)
- ③ All humans
 - Not realistic as this definition would neglect welfare of the citizens of other countries

Foreign-Income Tax Model

- Consumer utility: $u(x, y, y', z, n)$
- Tax policy of the domestic government: $x = c(y, z) + y'$
- Foreign Income is taxed at a higher marginal tax rate than domestic when:

$$\frac{\partial}{\partial n} \frac{u_y(x, y, z, n)}{u_z(x, y, z, n)} < 0$$

- note: case when untaxed foreign earnings y' are not present

Model, cont.

- When introducing y' , agent chooses value to maximize:

$$u(c(y, z) + y', y, y', z, n)$$

- Envelope theorem yields:

$$\bar{u}_y = u_y(c + y', y, y', z, n)$$

$$\bar{u}_z = u_z(c + y', y, y', z, n)$$

$$s(c, y, y', z, n) = \frac{u_y}{u_z}$$

such that foreign income is taxed higher at the margin if:

$$s_n + s_{y'} \frac{\partial y'}{\partial n} > 0$$

- note: s is willingness to substitute domestic and foreign earnings
- n and y' play opposition to each other on s , with $s_n < 0$ and $s_{y'} > 0$
- Implies that a lower marginal tax rate on domestic income is reasonable with y'

Connecting to Kleven et. al

- Use this type of model to show how migration patterns vary by ability level of footballers
- Takes exogenous policy changes as opportunity to test intuition provided by model
- More concerned with migration responses to the tax policy than defining optimal policy

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International Football, an Overview

- Football in Europe is a behemoth: 50+ leagues, 21,000 players
- Top Five Leagues: England, Spain, Germany, Italy, France
- €20.4bn in revenue from Top Five alone

Massive Player Transfer Market - €11bn in 2025

The Transfer Market

- Teams pay to purchase rights to players' services
- Two windows:
 - ① Summer: June - September 1, seen as main window
 - ② Winter: January
- System allows for player movement to be easily followed
- Contract numbers are commonly recorded, with approximations easily accessible

An Example of Transfers

- In 2025, Liverpool F.C. spent €520m during the summer breaking the Premier League record for most expensive transfer twice
- Importantly, not one of the seven incoming players are English or British
- In Liverpool's last match (at time of writing), they did not start a single British player, with only two senior British players being in the larger squad (20 players)
- How is this possible?

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What is Their Goal?

- How does variation in top tax rates affect the migration of footballers in Europe?
- Exploit variation across multiple dimensions:
 - ① Variation in top tax rates across countries and over time
 - ② Preferential taxation schemes in specific countries
 - ③ Bosman Rule
- Utilize a Mirrleesian theoretical framework applied to a multinomial logit to determine player choices

Policy Overview - Bosman

- Bosman ruling occurred in 1995, overhauling much of the football status quo
- Prior to Bosman ruling:
 - ① Three-player rule: no more than three foreign players per team could participate in any game in UEFA club competitions
 - ② Transfer-fee rule: require a transfer fee even when a player became a free agent
- Bosman posed as a major hurdle to mobility both domestically and across countries
- Note: Foreign-player rule was still in some sort of effect, applying to non-Europeans after 1995

Policy Overview - Top Tax Schema

Spain:

- Beckham law: Flat tax of 24% for foreign workers relocating to Spain after Jan. 1, 2004
- Reduction from 45% at time of passing
- Can not have been a taxable resident of Spain in any of the preceding 10 years to be eligible

Denmark:

- Researchers' Tax Scheme: Flat tax of 25% for high income foreigners signing for employment in Denmark after June 1, 1991
- Reduction from 68% top tax rate at time of introduction
- Can not have been tax liable to Denmark for 3 years prior to scheme and must make at least €103,000

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Baseline Model - Flexible Demand

- $N = \{n_1, n_2, \dots, c_N\}$ countries possessing a continuum of domestic potential players
- Each player is endowed with footballing ability $a \geq 0$
- If they play, the agent generates value a for their club
- As in Mirrlees, before-tax wage is equal to ability, a

Baseline Model, cont.

- Player also assigned country of origin m and preferences $\mu_m = (\mu_{1m}, \mu_{2m}, \dots, \mu_{Nm})$ for each location
- A player thus gets utility from country n by

$$u(a(1 - \tau_{nm})) + \mu_{nm}$$

- where $u(\cdot)$ is increasing and τ_{nm} is the tax rate in n for a player from m
- IC condition is thus

$$u(a(1 - \tau_{nm})) + \mu_{nm} \geq \max_{n'} \{u(a(1 - \tau_{n'm})) + \mu_{n'm}\}$$

Baseline Model, cont.

- Define $g_m(\mu_m, a)$ as the joint distribution of players from m with ability a playing in n
- This measure of player supply is denoted as p_{nma} henceforth
- Total Measure of players in country n from country m with any ability level as

$$p_{nm}(1 - \tau_{nm}) \equiv \int_0^{\infty} p_{nma}(a(1 - \tau_{nm}))da$$

Baseline Model, cont.

- Countries may separate tax rates such that $\tau_{nn} = \tau_{nd}$ and $\tau_{nm} = \tau_{nf} \forall n \neq m$
- Distributions by foreign and domestic can be defined as

Domestic:

$$p_{nd}(1 - \tau_{nd}) \equiv \int_0^{\infty} p_{nda}(a(1 - \tau_{nd}))da$$

Foreign:

$$p_{nf}(1 - \tau_{nf}) \equiv \int_0^{\infty} p_{nfa}(a(1 - \tau_{nf}))da$$

Baseline Model, cont.

Proposition 1:

Assuming that density $g_m(\mu_m, a)$ is smooth and positive everywhere on domain D , $p_{nda}, p_{nfa} > 0$ for all n, a and

- ① p_{nda} is decreasing in τ_{nd} and unaffected by τ_{nf}
- ② p_{nfa} is decreasing in τ_{nf} and unaffected by τ_{nd}

- States own-tax effect on number of both types of player is negative at all abilities
- Cross-tax effect is zero

Baseline Model, cont.

- Denote elasticities of players in a country n w.r.t. net-of-tax rate on players in country n

Foreign:

$$\varepsilon_{nf} = \frac{dp_{nf}}{d(1 - \tau_{nf})} \frac{1 - \tau_{nf}}{p_{nf}}$$

Domestic:

$$\varepsilon_{nd} = \frac{dp_{nd}}{d(1 - \tau_{nd})} \frac{1 - \tau_{nd}}{p_{nd}}$$

- Given approx. 85-90 percent of players are domestic, cutting rates on only foreign players can attract a large number of new players while having meager effects on preexisting foreign players

Adding Rigid Demand

Assumptions

- Assume each country's football market hires a continuum of measure one of players
- Players are hired by a continuum of clubs of measure one (each club hires one player)
- No new entry by clubs
- Population of potential native players in a country n has measure $P_n > 1$, meaning not all players can play in equilibrium
- Non-players work in a regular labor market with wage set to zero

Rigid Demand, cont.

Lemma 1:

In any equilibrium, within any country n , surplus $s_n \geq 0$ captured by each club is constant across all clubs and players in said country. Thus, before-tax wage paid out to a player of ability a in country n is $a - s_n$. No player with $a < s_n$ plays in country n .

- Form follows as before, except for before-tax income now being $a - s_n$

Rigid Demand, cont.

- Total number of players in a country n across ability levels is now represented as

Domestic:

$$p_{nd}(s_n, 1 - \tau_{nd}) \equiv \int_{s_n}^{\infty} p_{nda}((a - s_n)(1 - \tau_{nd}))$$

Foreign:

$$p_{nf}(s_n, 1 - \tau_{nf}) \equiv \int_{s_n}^{\infty} p_{nfa}((a - s_n)(1 - \tau_{nf}))$$

- Both functions decrease in s_n , but p_{ni} is increasing in $(1 - \tau_{ni})$ for $i = (f, d)$

Rigid Demand, cont.

- Only general equilibrium differs, requiring

$$\begin{aligned} p_{nd}(s_n, 1 - \tau_{nd}) + p_{nf}(s_n, 1 - \tau_{nf}) &= 1 \\ \Rightarrow s_n &= s_n(1 - \tau_{nd}, 1 - \tau_{nf}) \end{aligned}$$

- By inserting equilibrium surplus into $p_{nd}, p_{nda}, p_{nf}, p_{nfa}$, obtain relationships which are functions of tax rates

Rigid Demand, cont.

Proposition 2:

Assume that countries are small and that density $g_m(\mu_m, a)$ is smooth and positive everywhere on its domain D . Then, $p_{nda}, p_{nfa} > 0$ for all $a > s_n$, and:

- ① $s_n(1 - \tau_{nd}, 1 - \tau_{nf})$ decreases with τ_{nd} and τ_{nf}
- ② $p_{nda}(1 - \tau_{nd}, 1 - \tau_{nf})$ decreases with τ_{nd} at high abilities, increases with τ_{nd} at low abilities, and increases with τ_{nf} at all abilities
- ③ $p_{nfa}(1 - \tau_{nd}, 1 - \tau_{nf})$ decreases with τ_{nf} at high abilities, increases with τ_{nf} at low abilities, and increases with τ_{nd} at all abilities
- ④ $p_{nd}(1 - \tau_{nd}, 1 - \tau_{nf})$ decreases with τ_{nd} and increases with τ_{nf}
- ⑤ $p_{nf}(1 - \tau_{nd}, 1 - \tau_{nf})$ decreases with τ_{nf} and increases with τ_{nd}

Discussion of Model

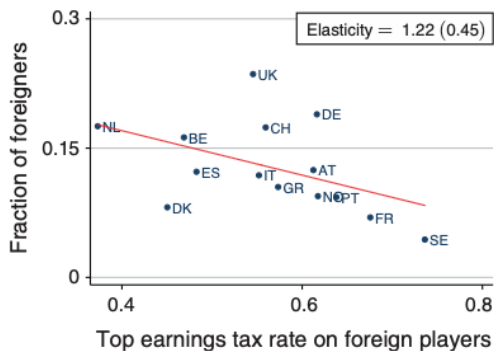
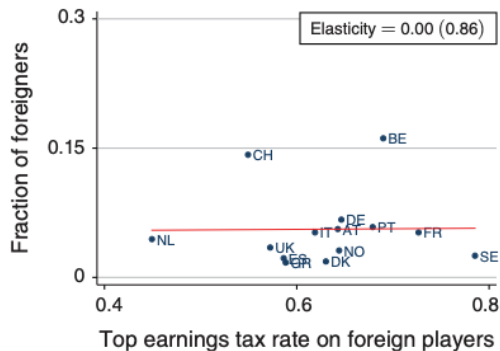
- Sorting effects related to own-tax and cross-tax effects:
 - ① Taxing foreign players no reduces the number of foreign players at all ability levels
 - ② Due to equilibrium sorting, there is now a cross-tax effect of taxing one group on the other
- High ability players have $a \gg s_n$ such that $a - s_n \simeq a$ (response to taxation is basically the same in both specifications), thus mobility elasticity for high-ability players is the upper bound relevant for other high-income occupations with flexible labor demand

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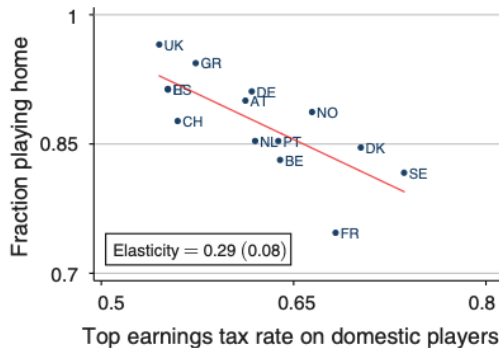
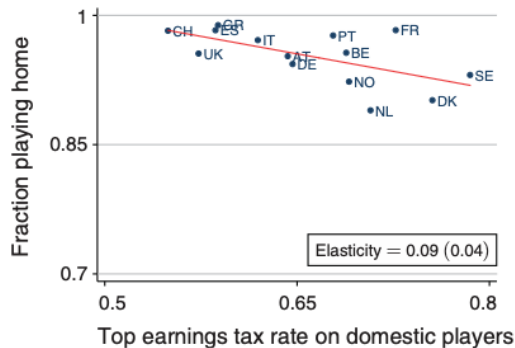
Bosman Ruling Cross-Country Correlations

Panel A. In-migration of foreign players



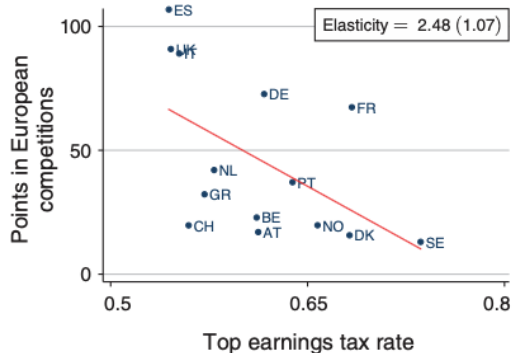
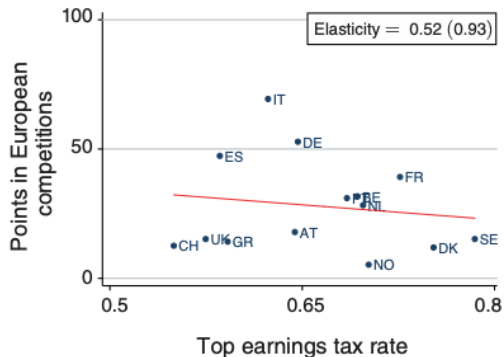
Bosman Ruling Cross-Country Correlations

Panel B. Out-migration of domestic players



Bosman Ruling Cross-Country Correlations

Panel C. Club performance



Reduced-form elasticity estimates

	Elasticity w.r.t. $(1 - \text{MTR})$ (1)	Elasticity w.r.t. $(1 - \text{ATR})$ (2)
Panel A. Bosman ruling		
<i>A1. Foreign players</i>		
Pre-Bosman	0.00 (0.86)	—
Post-Bosman	1.22 (0.45)	1.79 (0.50)
<i>A2. Domestic players</i>		
Pre-Bosman	0.09 (0.04)	—
Post-Bosman	0.29 (0.08)	0.27 (0.08)
Panel B. Country case studies		
<i>B1. Spain 2004 "Beckham law"</i>		
Top-quality players	1.49 (0.34)	1.87 (0.75)
Lower-quality players	1.43 (0.88)	1.62 (0.82)
Eligible players	1.29 (0.22)	1.41 (0.31)
Non-eligible (placebo)	0.57 (0.49)	0.45 (0.78)
<i>B2. Danish tax scheme for foreigners</i>		
Top-quality players	3.01 (0.78)	—
Lower-quality players	-0.70 (0.54)	—

Tax Reform Case Studies

- Compare treated country, Spain, to synthetic control created by a weighted average of the characteristics of other European countries
- Use a 2SLS approach to estimate elasticities:

$$\log(P_{ct}) = e \log(1 - \tau_{ct}) + \beta \mathbf{1}(c = T) + \gamma \mathbf{1}(t \geq t_0) + \varepsilon_{ct}$$

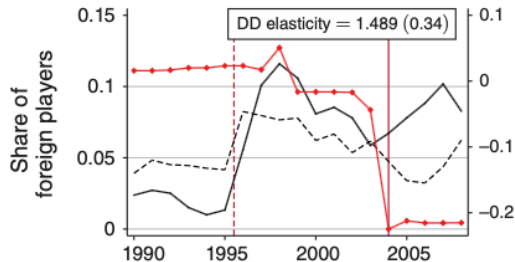
- Instrumented with:

$$\mathbf{1}(c = T) \times \mathbf{1}(t \geq t_0)$$

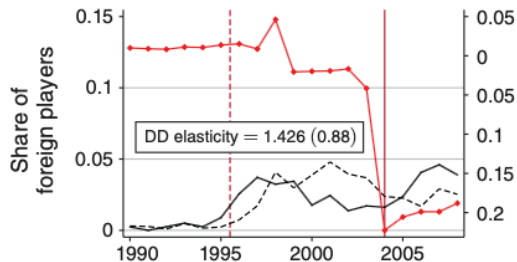
- Estimates capture medium-term responses to reform

Beckham Law Effects

Panel A1. Top-quality players



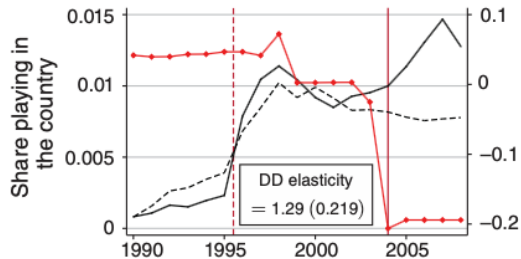
Panel A2. Lower-quality players



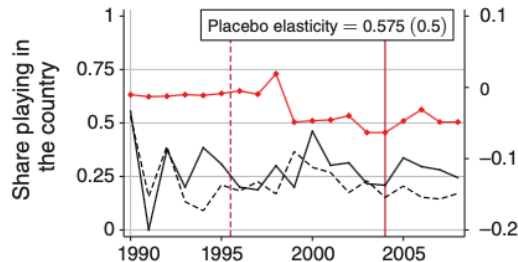
- Solid: Spain
- Dashed: Synthetic
- Red: Δ top tax rate

Beckham Law Effects, cont.

Panel B1. Eligible foreign players



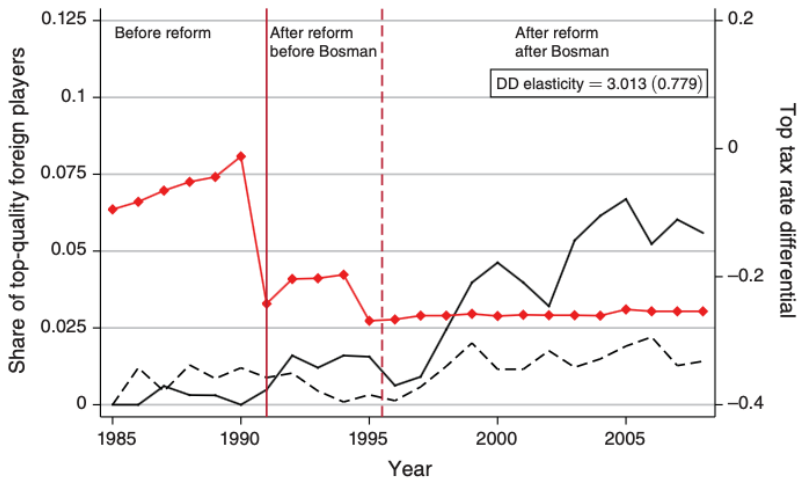
Panel B2. Non-eligible foreign players



- Solid: Spain
- Dashed: Synthetic
- Red: Δ top tax rate

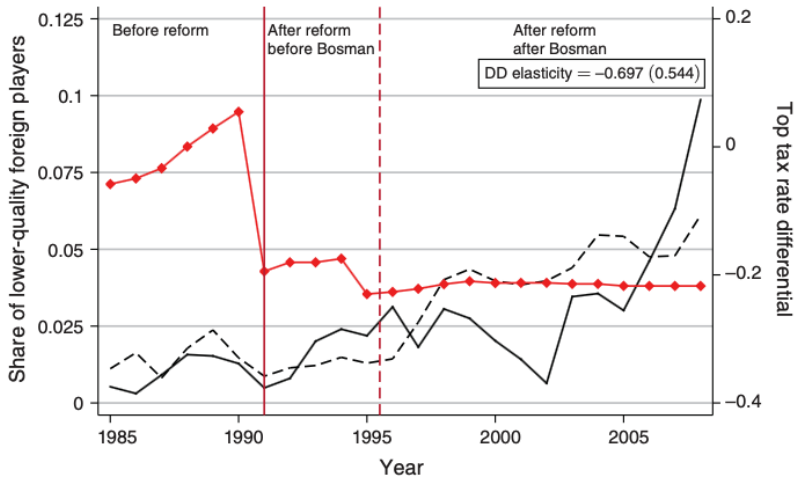
Danish Reform

Panel A. Top-quality players



Danish Reform, cont.

Panel B. Lower-quality quality players



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Empirical Multinomial Discrete Choice

$$\begin{aligned}
 U_{nt}^i &= u((1 - \tau_{nt}^i)w_{nt}^i) + \mu_{nt}^i \\
 &= \alpha \log(1 - \tau_{nt}^i) + \alpha \log(w_{nt}^i) + home_n^i + \mathbf{x}_t^i \beta_n + \gamma_n + \nu_{nt}^i
 \end{aligned}$$

- τ_{nt}^i is average tax rate on player i in country n at time t
- w_{nt}^i is before-tax wage
- μ_{nt}^i idiosyncratic preference for country n at time t comprised of:
 - ① preference for home country $home_n^i$, equal to one if n is the native country for player i
 - ② individual characteristics $\mathbf{x}_t^i$ like age, age-squared. Can vary by country
 - ③ unobservable country specific characteristics γ_n

Empirical, Cont.

- Allow ν_{nt}^i to be T1 EV $\rightarrow$ estimate Multinomial Logit
- Estimate first with only net-of-tax rate and home dummy as explanatory variables
- Add individual and country F.E.'s extension
- Then, contend with how to control for wage variation by:
 - ① Control for variation through ability channel, such that $w_{nt}^i = a_t^i$
 - ② $w_{nt}^i = a_t^i w_{nt}$ where w_{nt} captures trends in the overall market as well as cross country differences
 - ③ Control via country $\times$ year $\times$ ability F.E. in case of substitution between ability levels not being satisfied

Elasticity

- α is indicative of mobility responses to net-of-tax rate
 - $\alpha > 0$ implies a rise in n-o-t rate leads to a positive effect on the probability of a player locating in the country
- Derive probability of player i locating in country n w.r.t n-o-t rate in country n

Deriving Elasticity

$$P_{nt}^i = \frac{\exp\{\alpha \log(1 - \tau_{nt}^i) + \mathbf{X}_{\mathbf{nt}}^i \gamma\}}{\sum_m \exp\{\alpha \log(1 - \tau_{mt}^i) + \mathbf{X}_{\mathbf{mt}}^i \gamma\}}$$
$$\varepsilon_{nt}^i \equiv \frac{d \log(P_{nt}^i)}{d \log(1 - \tau_{nt}^i)} = \alpha \cdot (1 - P_{nt}^i)$$

- ε_{nt}^i is individual elasticity of P_{nt}^i w.r.t. $1 - \tau_{nt}^i$

Deriving Elasticity, cont.

Domestic Elasticity:

$$\varepsilon_{domestic}^n = \frac{\sum_{i \in I_n} \alpha \cdot (1 - P_{nt}^i) P_{nt}^i}{\sum_{i \in I_n} P_{nt}^i} = \alpha \cdot (1 - \bar{P}_n^d)$$

Foreign Elasticity:

$$\begin{aligned} \varepsilon_{foreign}^n &= \frac{\sum_{i \in I_n^C} dP_{nt}^i / d\log(1 - \tau_{nf})}{\sum_{i \in I_n^C} P_{nt}^i} \\ &= \frac{\sum_{i \in I_n^C} \alpha \cdot (1 - P_{nt}^i) P_{nt}^i}{\sum_{i \in I_n^C} P_{nt}^i} = \alpha \cdot (1 - \bar{P}_n^f) \end{aligned}$$

Empirical Results

	Post-Bosman (1996–2008)					Pre-Bosman	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Specifications with top marginal tax rates							
Utility parameter estimates							
$\log(1 - \text{MTR})$	1.323*** (0.073)	0.729*** (0.116)	1.089*** (0.159)	0.634*** (0.132)	1.034*** (0.156)	1.311*** (0.139)	0.202 (0.160)
Implied elasticities							
$\varepsilon_{\text{domestic}}$	0.156 (0.009)	0.074 (0.012)	0.121 (0.018)	0.070 (0.015)	0.112 (0.017)	0.089 (0.009)	0.011 (0.009)
$\varepsilon_{\text{foreigner}}$	1.308 (0.072)	0.704 (0.112)	1.057 (0.154)	0.621 (0.130)	0.991 (0.150)	0.747 (0.079)	0.199 (0.158)

Empirical Results, Cont.

	Post-Bosman (1996–2008)				
	(1)	(2)	(3)	(4)	(5)
Panel B. Specifications with grouped average tax rates					
Utility parameter estimates					
$\log(1 - \text{ATR})$	1.599*** (0.079)	0.931*** (0.138)	1.721*** (0.197)	1.123*** (0.161)	1.772*** (0.192)
Implied elasticities					
$\varepsilon_{\text{domestic}}$	0.184 (0.009)	0.093 (0.014)	0.184 (0.021)	0.122 (0.017)	0.190 (0.021)
$\varepsilon_{\text{foreigner}}$	1.582 (0.078)	0.900 (0.133)	1.654 (0.190)	1.100 (0.157)	1.698 (0.184)
Country F-E	No	Yes	Yes	Yes	Yes
Age, age squared, exp., quality $\times$ country F-E	No	Yes	Yes	Yes	Yes
Year $\times$ country F-E	No	No	Yes	Yes	Yes
Age, age squared, exp., quality $\times$ year $\times$ country F-E	No	No	No	Yes	No
Observations	55,225	55,225	55,225	55,225	55,225

Sorting Effects, Cross-Effects, and Second Leagues (1/2)

	(1)	(2)	(3)	(4)	(5)
Panel A. Top leagues (1999–2008)					
$\log(1 - \tau)$	1.795*** (0.065)	1.138*** (0.118)	1.433*** (0.155)		
$\log(1 - \tau) \times low$				-0.512*** (0.149)	-0.529*** (0.145)
$\log(1 - \tau) \times top$				1.409*** (0.136)	1.301*** (0.132)
$\log(1 - \tau^f) \times domestic$					-0.618*** (0.119)
$\log(1 - \tau^d) \times foreign$					-0.149 (0.178)

Sorting Effects, Cross-Effects, and Second Leagues (2/2)

	(1)	(2)	(3)	(4)	(5)
Panel B. Adding the five best second leagues (1999–2008)					
$\log(1 - \tau)$	1.334*** (0.077)	0.995*** (0.128)	1.226*** (0.169)		
$\log(1 - \tau) \times low$				-0.391* (0.158)	-0.448** (0.154)
$\log(1 - \tau) \times top$				1.494*** (0.136)	1.409*** (0.133)
$\log(1 - \tau^f) \times domestic$					-0.635*** (0.134)
$\log(1 - \tau^d) \times foreign$					-0.201 (0.192)
Country F-E	No	Yes	Yes	Yes	Yes
Age, age squared, exp., quality $\times$ country F-E	No	Yes	Yes	Yes	Yes
Year $\times$ country F-E	No	No	Yes	No	No

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Conclusions and Takeaways

- Mobility responses to top tax rate changes is large
- Elasticity on foreign players w.r.t. net-of-tax rate is about 1, even larger for top-quality foreign players
- Cutting taxes on foreign and local is less cost-effective than only cutting for foreign players
- Location elasticities are largest at the top[and negative at the bottom, reflecting ability sorting
- Provide upper bound on migration response of labor market

Next Steps

- Take framework to analyze migrate and return efficacy through football loans
- Potentially applicable to human capital development → overall development as many loan players are young and looking to develop skill
- Utilize many data sources from this paper to get effects for footballers going on loan to get the opportunity to develop